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Vehicle fault prediction

Abstract

A computer includes a processor and a memory, and the memory stores instructions executable by the processor to receive time-series data from sensors of a vehicle, determine an efficiency of the vehicle, and determine a probability of a fault occurring in the vehicle based on the time-series data and on the efficiency. The efficiency is energy consumption by the vehicle per distance traveled by the vehicle.

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Background/Summary

BACKGROUND

(1) Modern vehicles typically include on-board diagnostics, which are systems to self-diagnose and report issues on board a vehicle. Reporting issues may follow a standard such as On-Board Diagnostics II (OBD-II). The vehicle may output a message indicating an issue through a physical port. The OBD-II standard specifies a 16-pin D-shaped female connector as the port for outputting messages. The message may take the form of a diagnostic trouble code (DTC). According to OBD-II, the format of a DTC is a letter followed by four digits, the combination of which can identify a particular issue with the vehicle, e.g., a fault associated with a vehicle component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of an example vehicle.
- (2) FIG. 2 is a diagram of a machine-learning program trained to predict a fault in the vehicle.
- (3) FIG. 3 is a process flow diagram of an example process for predicting the fault in the vehicle.

DETAILED DESCRIPTION

(4) This disclosure describes techniques for predicting faults on board a vehicle before those faults occur, e.g., before an OBD-II system of the vehicle would otherwise output a diagnostic trouble code. A computer is programmed to receive time-series data from sensors of the vehicle, determine an efficiency of the vehicle, and determine a probability of a fault occurring in the vehicle based on the time-series data and on the efficiency. A measurement of the efficiency of the vehicle is energy consumption by the vehicle per distance traveled by the vehicle. In some circumstances, inefficient operation of the vehicle may indicate an impending fault, permitting the impending fault to be addressed before becoming an actual fault. The time-series data from the sensors may be values that affect the efficiency, e.g., internal temperatures, tire pressures, etc. The use of the time-series data may permit the computer to identify the circumstances in which inefficient operation indicates an impending fault, as well as which fault is impending. A fault or possible fault may thus be identified earlier than it would have been otherwise.

(5) A computer includes a processor and a memory, and the memory stores instructions executable by the processor to receive time-series data from sensors of a vehicle, determine an efficiency of the vehicle, and determine a probability of a fault occurring in the vehicle based on the time-series data and on the efficiency. The efficiency being energy consumption by the vehicle per distance traveled by the vehicle.

(6) In an example, the instructions may further include instructions to output a message in response to the probability exceeding a threshold.

(7) In an example, the instructions may further include instructions to determine a plurality of probabilities including the probability of a plurality of respective faults including the fault occurring in the vehicle. In a further example, the instructions may further include instructions to output a message in response to at least one of the probabilities exceeding a threshold.

(8) In another further example, the memory may store a plurality of thresholds for the respective faults, the thresholds may include at least two different values, and the instructions may further include instructions to output a message in response to at least one of the probabilities exceeding the respective threshold.

(9) In an example, the instructions may further include instructions to determine the probability in response to an indication of a completion of a trip of the vehicle. In a further example, the indication may be one of turning off the vehicle or shifting the vehicle to park.

(10) In another further example, the instructions may further include instructions to determine a trip-level metric either applicable to the trip as a whole or indicating a state of the vehicle at a beginning and the completion of the trip, and determining the probability may be based on the trip-level metric.

(11) In an example, the instructions may further include instructions to determine the probability by executing a machine-learning program, and the time-series data and the efficiency are inputs to the machine-learning program. In a further example, the machine-learning program may include a feature-extraction program, and the time-series data may be an input to the feature-extraction program. In a yet further example, the machine-learning program may include a classification program, and an output of the feature-extraction program may be an input to the classification program. In a still yet further example, the efficiency may be an input directly to the classification program.

(12) In another still yet further example, the instructions may further include instructions to determine a trip-level metric either applicable to a trip of the vehicle as a whole or indicating a state

of the vehicle at a beginning and at a completion of the trip, and the trip-level metric may be an input directly to the classification program.

(13) In another further example, the machine-learning program may be trained on ground-truth faults and training data, the training data may include training time-series data and training efficiencies, and the training data may be paired to the ground-truth faults. In a yet further example, the ground-truth faults may be diagnostic trouble codes.

(14) In an example, the time-series data may include a temperature of a vehicle component.

(15) In an example, the time-series data may include a tire pressure of a tire of the vehicle.

(16) In an example, the fault may correspond to a diagnostic trouble code.

(17) In an example, the instructions may further include instructions to at least one of output a message to a screen of the vehicle, transmit the message to a mobile device, or transmit the message to a server remote from the vehicle.

(18) A method includes receiving time-series data from sensors of a vehicle, determining an efficiency of the vehicle, and determining a probability of a fault occurring in the vehicle based on the time-series data and on the efficiency. The efficiency is energy consumption by the vehicle per distance traveled by the vehicle.

(19) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a vehicle computer **105** or a remote computer **110** includes a processor and a memory, and the memory stores instructions executable by the processor to receive time-series data **205** from sensors **115** of a vehicle **100**, determine an efficiency **210** of the vehicle **100**, and determine a probability **215** of a fault occurring in the vehicle **100** based on the time-series data **205** and on the efficiency **210**. The efficiency **210** is energy consumption by the vehicle **100** per distance traveled by the vehicle **100**.

(20) With reference to FIG. **1**, the vehicle **100** may be any passenger or commercial automobile such as a car, a truck, a sport utility vehicle, a crossover, a van, a minivan, a taxi, a bus, etc.

(21) The vehicle **100** includes the vehicle computer **105**. The vehicle computer **105** is a microprocessor-based computing device, e.g., a generic computing device including a processor and a memory, an electronic controller or the like, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a combination of the foregoing, etc. Typically, a hardware description language such as VHDL (VHSIC (Very High Speed Integrated Circuit) Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. The vehicle computer **105** can thus include a processor, a memory, etc. The memory of the vehicle computer **105** can include media for storing instructions executable by the processor as well as for electronically storing data and/or databases, and/or the vehicle computer **105** can include structures such as the foregoing by which programming is provided. The vehicle computer **105** can be multiple computers coupled together.

(22) The vehicle computer **105** may transmit and receive data on board the vehicle **100** through a communications network **120** such as a controller area network (CAN) bus, Ethernet, WiFi, Local Interconnect Network (LIN), onboard diagnostics connector (OBD-II), and/or by any other wired or wireless communications network. The vehicle computer **105** may be communicatively coupled to the sensors **115**, a user interface **125**, a transceiver **130**, and other components of the vehicle **100** via the communications network **120**.

(23) The sensors **115** may provide data about operation of the vehicle **100**, for example, wheel speed, wheel orientation, tire pressure, and engine and transmission data (e.g., temperature, fuel consumption, etc.). The sensors **115** may detect the location and/or orientation of the vehicle **100**. For example, the sensors **115** may include global positioning system (GPS) sensors; accelerometers such as piezo-electric or microelectromechanical systems (MEMS); gyroscopes such as rate, ring

laser, or fiber-optic gyroscopes; inertial measurements units (IMU); and magnetometers. The sensors **115** may detect the external world, e.g., objects and/or characteristics of surroundings of the vehicle **100**, such as other vehicles, road lane markings, traffic lights and/or signs, pedestrians, etc. For example, the sensors **115** may include radar sensors, ultrasonic sensors, scanning laser range finders, light detection and ranging (lidar) devices, and image processing sensors such as cameras. (24) The user interface **125** presents information to and receives information from an occupant of the vehicle **100**. The user interface **125** may be located, e.g., on an instrument panel in a passenger cabin of the vehicle **100**, or wherever may be readily seen by the occupant. The user interface **125** may include dials, digital readouts, screens, speakers, and so on for providing information to the occupant, e.g., human-machine interface (HMI) elements such as are known. The user interface **125** may include buttons, knobs, keypads, microphone, and so on for receiving information from the occupant.

(25) The transceiver **130** may be adapted to transmit signals wirelessly through any suitable wireless communication protocol, such as cellular, Bluetooth®, Bluetooth® Low Energy (BLE), ultra-wideband (UWB), WiFi, IEEE 802.11a/b/g/p, cellular-V2X (CV2X), Dedicated Short-Range Communications (DSRC), other RF (radio frequency) communications, etc. The transceiver **130** may be adapted to communicate with a remote server, that is, a server distinct and spaced from the vehicle **100**. The remote server may be located outside the vehicle **100**. For example, the remote server may be associated with another vehicle (e.g., V2V communications), an infrastructure component (e.g., V2I communications), an emergency responder, etc. The remote server may be the remote computer **110**, a mobile device **135**, e.g., associated with the operator or owner of the vehicle **100**, etc. The transceiver **130** may be one device or may include a separate transmitter and receiver.

(26) The remote computer **110** is a microprocessor-based computing device, e.g., a generic computing device including a processor and a memory. The memory of the remote computer **110** can include media for storing instructions executable by the processor as well as for electronically storing data and/or databases, and/or the remote computer **110** can include structures such as the foregoing by which programming is provided. The remote computer **110** can be multiple computers coupled together.

(27) The mobile device **135** is a portable computing device such as a mobile phone, e.g., a smartphone, or a tablet. The mobile device **135** is a computing device including a processor and a memory. The mobile device **135** is owned and carried by a person who may be the operator or owner of the vehicle **100**.

(28) For the purposes of this disclosure, a “fault” is an issue that detracts from the operation of the vehicle **100**. The faults may correspond to respective diagnostic trouble codes, either one-to-one or to groups of related diagnostic trouble codes. A diagnostic trouble code is a sequence of characters identifying a particular type of fault. For example, diagnostic trouble codes under the OBD-II standard are five characters long. The first character is one of four letters: “P” for powertrain, “C” for chassis, “B” for body, and “U” for network. The second character is either 0 (zero) for a generic code or a 1 (one) for a manufacturer-specific code. The third character is a digit identifying a subsystem in which the fault is located. The fourth and fifth characters are a two-digit number serving as an index of the specific fault.

(29) The vehicle computer **105** or remote computer **110** may be programmed to determine one or more probabilities **215** of one or more faults occurring in the vehicle **100** in response to an indication of a completion of a trip of the vehicle **100**. For the purposes of this disclosure, a “trip” is defined as the vehicle **100** moving from a particular state in a first location to a particular state in a second location. The first and second locations may be different; i.e., the vehicle **100** has traveled. The states in the two locations may be the same. For example, the state may be that the vehicle **100** is off or that the vehicle **100** is in park. The indication may be an event suggesting that a trip has completed, e.g., turning off the vehicle **100** when the vehicle **100** is on, shifting the vehicle **100**

from another gear to park, etc.

(30) With reference to FIG. 2, the vehicle computer **105** or remote computer **110** may be programmed to receive the time-series data **205** from the sensors **115**. As will be generally understood, and for purposes of this disclosure, “time-series data” are values of one or more variables at discrete successive points of time. The time-series data **205** may be chosen that affect the efficiency **210** of the vehicle **100**. For example, the time-series data **205** may include one or more temperatures of one or more respective components of the vehicle **100**, e.g., a transmission, engine oil, etc. For another example, the time-series data **205** may include one or more tire pressures of one or more respective tires of the vehicle **100**. For another example, the time-series data **205** may include variables describing the motion of the vehicle **100**, e.g., speed, torque, yaw rate, acceleration, etc. For another example, the time-series data **205** may be environmental data such as ambient temperature, precipitation, etc. The vehicle computer **105** may receive the time-series data **205** over the communications network **120**, e.g., in real time. The remote computer **110** may receive the time-series data **205** via the transceiver **130**, e.g., in a single transmission in response to the indication of the completion of the trip of the vehicle **100**.

(31) The vehicle computer **105** or remote computer **110** may be programmed to determine one or more trip-level metrics **220**. A trip-level metric **220** is either applicable to the trip as a whole or indicates a state of the vehicle **100** at a beginning and the completion of the trip. The trip-level metrics **220** may be chosen that affect the efficiency **210** of the vehicle **100**. For example, one trip-level metric **220** may be a reduction in charge stored by batteries, e.g., high-voltage batteries, of the vehicle **100** if the vehicle **100** is a battery-electric vehicle. The reduction in charge is the difference between the charge at the beginning of the trip and the charge at the end of the trip. For another example, one trip-level metric **220** may be a volume of fuel consumed during the trip if the vehicle **100** is an internal-combustion vehicle. For another example, one trip-level metric **220** may be a distance traveled during the trip, e.g., as measured by an odometer of the vehicle **100**. For another example, one trip-level metric **220** may be a duration of the trip, i.e., a difference between an ending time of the trip and a starting time of the trip. For another example, one trip-level metric **220** may be a change in altitude from the beginning of the trip to the completion of the trip. For another example, one trip-level metric **220** may be a beginning location and ending location of the trip.

(32) The vehicle computer **105** or remote computer **110** may be programmed to determine an efficiency **210** of the vehicle **100**. The efficiency **210** is energy consumption by the vehicle **100** per distance traveled by the vehicle **100**. For example, the computer may determine the efficiency **210** from the trip-level metrics **220**, e.g., the reduction in charge stored by the batteries divided by the distance traveled during the trip.

(33) The vehicle computer **105** or remote computer **110** may be programmed to determine the one or more probabilities **215** of the one or more faults occurring in the vehicle **100** based on the time-series data **205**, the efficiency **210**, and/or the trip-level metrics **220**. Each probability **215** is a likelihood of the respective fault occurring in the vehicle **100**, e.g., within a specific time or a within a particular distance traveled by the vehicle **100**. A fault occurring in the vehicle **100** may be indicated by the vehicle computer **105** outputting the diagnostic trouble code(s) corresponding to the fault. The probabilities **215** may also include a probability **215** of none of the faults occurring in the vehicle **100**.

(34) For example, the vehicle computer **105** or remote computer **110** may be programmed to determine the probabilities **215** by executing an expert system. The expert system may be a deterministic, rule-based system. The rules may be chosen based on, e.g., heuristics used by individuals involved in designing electric vehicles. The inputs to the expert system may include the time-series data **205**, the efficiency **210**, and/or the trip-level metrics **220**.

(35) For another example, the vehicle computer **105** or remote computer **110** may be programmed to determine the probabilities **215** by executing a machine-learning program **200**. The inputs to the

machine-learning program **200** may include the time-series data **205**, the efficiency **210**, and/or the trip-level metrics **220**. The outputs of the machine-learning program **200** may include at least one probability **215** of a fault occurring in the vehicle **100**, e.g., a plurality of probabilities **215** of a plurality of respective faults occurring in the vehicle **100**.

(36) The machine-learning program **200** may include a feature-extraction program **225** and a classification program **230**. The time-series data **205** may be an input to the feature-extraction program **225**. An output of the feature-extraction program **225**, which will be referred to as intermediate data **235**, may be an input directly to the classification program **230**. The efficiency **210** and the one or more trip-level metrics **220** may also be inputs directly to the classification program **230**. The probabilities **215** may be outputs of the classification program **230**.

(37) The feature-extraction program **225** may be programmed to identify features in the time-series data **205** and output the features as the intermediate data **235**. For the purposes of this disclosure, a “feature” of the time-series data **205** is a piece of information about the content of the time-series data **205**, specifically about whether a certain range of the time-series data **205** has certain properties, e.g., a spike, a constant value, a repeating periodic pattern, etc. The intermediate data **235** may have a reduced dimensionality compared to the time-series data **205**. For example, the feature-extraction program **225** may be a convolutional neural network. The convolutional neural network may include, e.g., convolutional layers to find features within time windows over the time-series data **205**, attention layers to find features with dependencies across larger time windows, recurrent layers to find features with dependencies in different time windows, etc. For another example, the feature-extraction program **225** may be an encoder. The encoder may be or include an artificial neural network such as a residual neural network (ResNet). The encoder may include a sequence of layers and may reduce a dimensionality of the data from one layer to the next layer. The output of the final layer, i.e., the intermediate data **235**, may be a latent vector.

(38) The classification program **230** may be programmed to receive the efficiency **210**, the trip-level metrics **220**, and the intermediate data **235** as inputs and to classify the situation defined by the inputs as one of the faults or as no fault. Each probability **215** outputted by the classification program **230** represents a respective strength of the classification of the situation as the respective fault or no fault. For example, the classification may be a support vector machine. For another example, the classification program **230** may be an ensemble model with a plurality of “weak learners,” e.g., logistic regression, decision trees, etc. For another example, the classification program **230** may be a naïve Bayes classifier.

(39) The machine-learning program **200** may be trained on ground-truth faults and training data. The training data may include training time-series data and training efficiencies. The training data are paired to the ground-truth faults. The machine-learning program **200** may be trained to replicate the ground-truth faults when given the training data as inputs. The ground-truth faults may be diagnostic trouble codes, e.g., the diagnostic trouble codes corresponding to the ground-truth faults as described above. The ground-truth faults and training data may be historical data gathered from vehicles.

(40) The machine-learning program **200** may be trained using any suitable method for the type of machine learning chosen. For example, the feature-extraction program **225** may be trained using stochastic gradient descent and backpropagation. For another example, if the classification program **230** is an ensemble model, the classification program **230** may be trained using bagging (e.g., Random Forest), gradient boosting (e.g., boosted decision trees, boosted logistic regression), etc.

(41) The memory of the vehicle computer **105** or remote computer **110** may store a plurality of thresholds for the respective faults. The thresholds may include at least two different values, e.g., a unique value for each fault. The thresholds may be chosen based on calibrating a risk of false positives versus a consequence of the fault.

(42) The vehicle computer **105** or remote computer **110** may be programmed to actuate a component of the vehicle **100** in response to a probability **215** exceeding a threshold, e.g., at least

one of the probabilities **215** exceeding the respective threshold. For example, actuating the component may include outputting a message. The message may indicate the fault(s) for which the respective probability **215** exceeds the respective threshold, e.g., by including the corresponding diagnostic trouble code(s). For example, actuating the component may include at least one of instructing the user interface **125** to output the message to a screen of the vehicle **100**, transmitting the message to the mobile device **135**, e.g., via the transceiver **130**, or transmitting the message to a server remote from the vehicle **100**, e.g., via the transceiver **130** to the remote computer **110**. The server may be, e.g., associated with a service center or fleet operator of the vehicle **100**. An operator of the vehicle **100** or a technician may thus be alerted to the possibility of the fault occurring in the vehicle **100** and may have the opportunity to preemptively address the fault, e.g., by scheduling a repair, creating a repair plan, ordering parts, etc.

(43) For another example, actuating the component may include modifying operation of a propulsion system of the vehicle **100**. For example, the vehicle computer **105** may limit the acceleration, speed, and/or revolutions per minute (RPMs) of the vehicle **100** to maximum values. For another example, the vehicle computer **105** may switch the vehicle **100** to a different driving mode. A driving mode can specify power maps defining throttle response, suspensions stiffness, steering feel, traction control, etc. The vehicle computer **105** may switch to a driving mode that is less stressful on the vehicle **100**, e.g., from a sport mode or comfort mode to an eco mode. For another example, for a hybrid or battery-electric vehicle **100**, the vehicle computer **105** may turn off an electric motor associated with the probability **215** that exceeded the threshold, leaving an internal-combustion engine or another electric motor operating to accelerate the vehicle **100**.

(44) FIG. 3 is a process flow diagram illustrating an example process **300** for predicting the faults in the vehicle **100**. The memory of the vehicle computer **105** and/or remote computer **110** stores executable instructions for performing the steps of the process **300** and/or programming can be implemented in structures such as mentioned above. The process **300** may be performed entirely on the vehicle computer **105** or the remote computer **110** (except as described), or the vehicle computer **105** may begin performing the process **300** and transmit data partway through the process **300** to the remote computer **110** for the remote computer **110** to complete the process **300**. As a general overview of the process **300**, the vehicle computer **105** receives the time-series data **205** in real time. In response to an indication of a completion of a trip of the vehicle **100**, the vehicle computer **105** or the remote computer **110** determines the trip-level metrics **220**, determines the efficiency **210**, and determines the probabilities **215**. In response to at least one of the probabilities **215** exceeding the respective threshold, the vehicle computer **105** or remote computer **110** sets the diagnostic trouble code and actuates a component.

(45) The process **300** begins in a block **305**, in which the vehicle computer **105** receives the time-series data **205** from the sensors **115** over the communications network **120**, e.g., in real time, as described above.

(46) Next, in a decision block **310**, the vehicle computer **105** and/or remote computer **110** determines whether there is an indication of a completion of a trip of the vehicle **100**. The vehicle computer **105** determines whether the indication is present as described above. The remote computer **110** may determine whether the indication is present based on whether the remote computer **110** has received a message from the vehicle computer **105** via the transceiver **130** indicating that the trip is complete. The message may also include the time-series data **205** collected in the block **305**, so the remote computer **110** thereby receives the time-series data **205**. The message may also include the data necessary for determining the trip-level metrics **220** and/or the efficiency **210**. In response to the indication of the completion of the trip, the process **300** proceeds to a block **315**. Otherwise, the process **300** returns to the block **305** for the vehicle computer **105** to continue receiving new time-series data **205**.

(47) In the block **315**, the vehicle computer **105** or remote computer **110** determines the trip-level metrics **220**, as described above.

(48) Next, in a block **320**, the vehicle computer **105** or remote computer **110** determines the efficiency **210** of the vehicle **100**, as described above.

(49) Next, in a block **325**, the vehicle computer **105** or remote computer **110** determines the probabilities **215** of the faults and of no fault based on the time-series data **205** from the block **305**, the trip-level metrics **220** from the block **315**, and the efficiency **210** from the block **320**, as described above.

(50) Next, in a decision block **330**, the vehicle computer **105** or remote computer **110** determines whether at least one of the probabilities **215** determined in the block **325** exceeds the respective threshold, as described above. If none of the probabilities **215** exceed the respective thresholds, the process **300** ends. In response to at least one of the probabilities **215** exceeding the respective threshold, the process **300** proceeds to a block **335**.

(51) In the block **335**, the vehicle computer **105** or remote computer **110** sets the diagnostic trouble code(s) corresponding to the fault. The vehicle computer **105** may set a value in memory for the diagnostic trouble code(s) corresponding to the fault. The remote computer **110** may transmit an instruction to the vehicle computer **105** via the transceiver **130** to set the value in memory for the diagnostic trouble code(s) corresponding to the fault.

(52) Next, in a block **340**, the vehicle computer **105** or the remote computer **110** actuates a component, e.g., by outputting a message, as described above. After the block **340**, the process **300** ends.

(53) In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an on-board vehicle computer, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

(54) Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

(55) A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless

communication, including the internals that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

(56) Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a nonrelational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

(57) In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

(58) In the drawings, the same reference numbers indicate the same elements. Further, some or all of these elements could be changed. With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted.

(59) All terms used in the claims are intended to be given their plain and ordinary meanings as understood by those skilled in the art unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. Use of “in response to” and “upon determining” indicates a causal relationship, not merely a temporal relationship.

(60) The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A computer comprising a processor and a memory, the memory storing instructions executable by the processor to: receive time-series data from sensors of a vehicle; determine an efficiency of the vehicle, the efficiency being energy consumption by the vehicle per distance traveled by the vehicle; in response to an indication of a completion of a trip of the vehicle, determine a plurality of probabilities of a plurality of respective faults occurring in the vehicle based on the time-series data and on the efficiency, the memory storing a plurality of thresholds for the respective faults, the thresholds having different values dependent on respective types of the respective faults; modify operation of a propulsion system of the vehicle based on the probability of the fault; and output a message in response to at least one of the probabilities exceeding the respective threshold.
2. The computer of claim 1, wherein the indication is one of turning off the vehicle or shifting the vehicle to park.

3. The computer of claim 1, wherein the instructions further include instructions to determine a trip-level metric applicable only to the single trip as a whole, and determining the probabilities is based on the trip-level metric.
 4. The computer of claim 1, wherein the instructions further include instructions to determine the probabilities by executing a machine-learning program, and the time-series data and the efficiency are inputs to the machine-learning program.
 5. The computer of claim 4, wherein the machine-learning program includes a feature-extraction program, and the time-series data is an input to the feature-extraction program.
 6. The computer of claim 5, wherein the machine-learning program includes a classification program, and an output of the feature-extraction program is an input to the classification program.
 7. The computer of claim 6, wherein the efficiency is an input directly to the classification program.
 8. The computer of claim 6, wherein the instructions further include instructions to determine a trip-level metric either applicable only to the single trip of the vehicle as a whole or indicating a difference between a state of the vehicle at a beginning of the trip and at the completion of the same trip, and the trip-level metric is an input directly to the classification program.
 9. The computer of claim 4, wherein the machine-learning program is trained on ground-truth faults and training data, the training data including training time-series data and training efficiencies, the training data being paired to the ground-truth faults.
 10. The computer of claim 9, wherein the ground-truth faults are diagnostic trouble codes.
 11. The computer of claim 1, wherein the time-series data includes a temperature of a vehicle component.
 12. The computer of claim 1, wherein the time-series data includes a tire pressure of a tire of the vehicle.
 13. The computer of claim 1, wherein the types of the faults correspond to respective diagnostic trouble codes.
 14. The computer of claim 1, wherein the instructions to output the message include instructions to at least one of output the message to a screen of the vehicle, transmit the message to a mobile device, or transmit the message to a server remote from the vehicle.
 15. The computer of claim 1, wherein the instructions further include instructions to determine a trip-level metric indicating a difference between a state of the vehicle at a beginning of the trip and the completion of the same trip, and determining the probabilities is based on the trip-level metric.
 16. A method comprising: receiving time-series data from sensors of a vehicle; determining an efficiency of the vehicle, the efficiency being energy consumption by the vehicle per distance traveled by the vehicle; in response to an indication of a completion of a trip of the vehicle, determining a plurality of probabilities of a plurality of respective faults occurring in the vehicle based on the time-series data and on the efficiency, the memory storing a plurality of thresholds for the respective faults, the thresholds having different values dependent on respective types of the respective faults; modifying operation of a propulsion system of the vehicle based on the probability of the fault; and outputting a message in response to at least one of the probabilities exceeding the respective threshold.
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